

# Problem Set 3

## Applied Stats/Quant Methods 1

Due: November 13, 2025

### Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in R, please include the code you used to get your answers. Please also include the .R file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday November 13, 2025. No late assignments will be accepted.

In this problem set, you will run several regressions and create an add variable plot (see the lecture slides) in R using the `incumbents_subset.csv` dataset. Include all of your code.

### Question 1

We are interested in knowing how the difference in campaign spending between incumbent and challenger affects the incumbent's vote share.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `difflog`.

```
1 lm(voteshare ~ difflog, inc.sub)
```

Call:

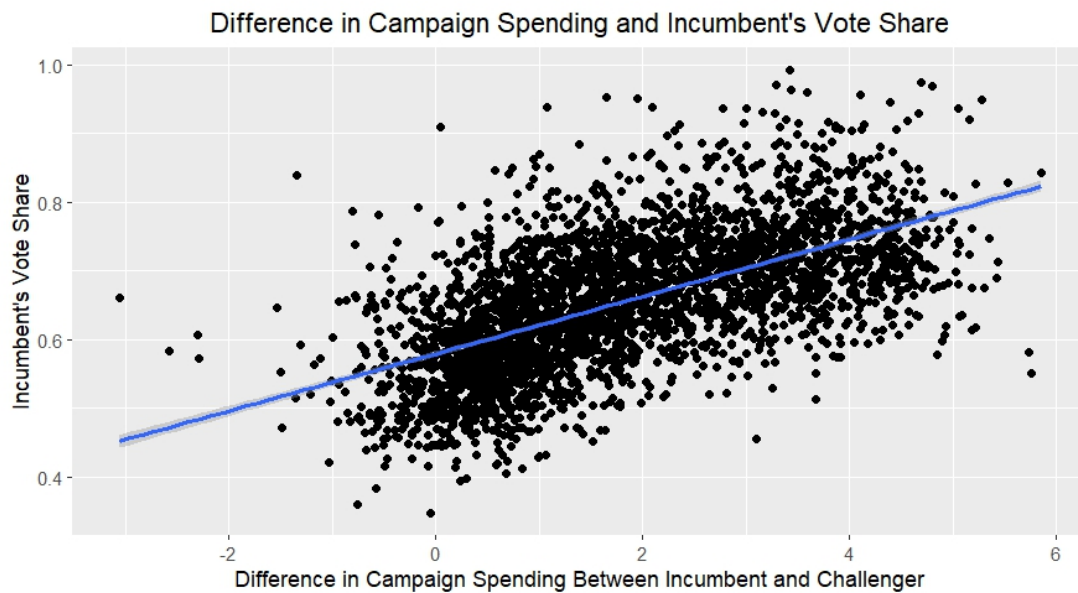
```
lm(formula = voteshare ~ difflog, data = inc.sub)
```

Coefficients:

|             |         |
|-------------|---------|
| (Intercept) | difflog |
| 0.57903     | 0.04167 |

2. Make a scatterplot of the two variables and add the regression line.

```
1 theme_update(plot.title = element_text(hjust = 0.5))
2
3 ggplot(inc.sub, aes(difflog, voteshare)) +
4   geom_point() +
5   labs(title = "Difference in Campaign Spending and Incumbent's Vote
6         Share",
7         x = "Difference in Campaign Spending Between Incumbent and
8         Challenger",
9         y = "Incumbent's Vote Share"
10  ) +
11  geom_smooth(method = lm)
```



3. Save the residuals of the model in a separate object.

```
1 regression1 <- lm(voteshare ~ difflog, inc.sub)
2 residuals1 <- regression1$residuals
```

4. Write the prediction equation.

The numbers representing the slope and y-intercept are taken from the linear regression run in R.

$$\hat{Y} = 0.04167X + 0.57903$$

In the equation, Y-hat represents the predicted value of the incumbent's vote share, and X represents the difference in campaign spending between incumbent and challenger.

## Question 2

We are interested in knowing how the difference between incumbent and challenger's spending and the vote share of the presidential candidate of the incumbent's party are related.

1. Run a regression where the outcome variable is `presvote` and the explanatory variable is `difflog`.

```
1 lm(presvote ~ difflog, inc.sub)
```

Call:

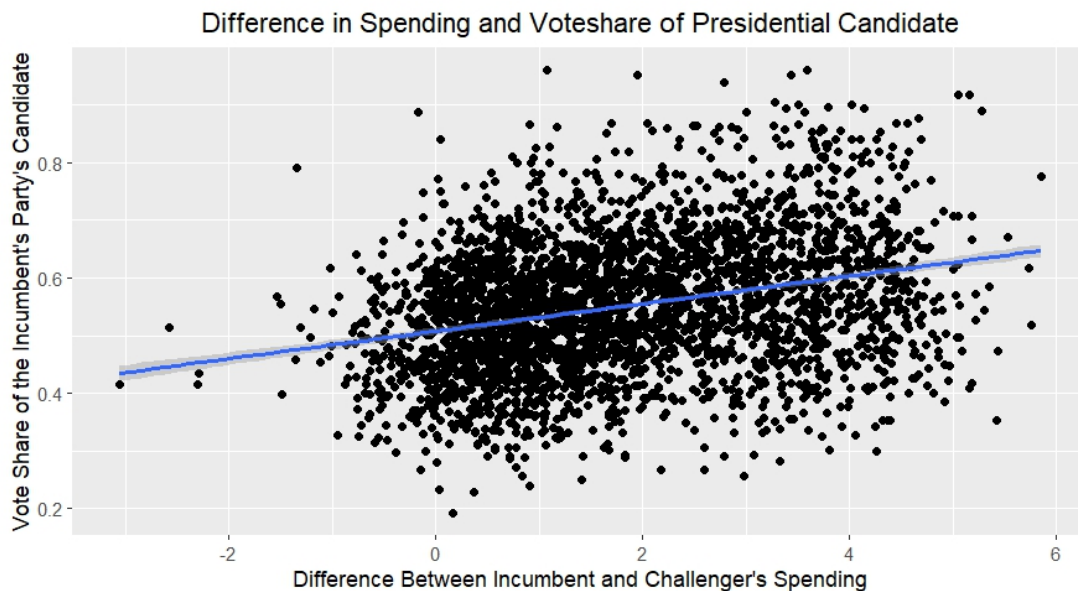
```
lm(formula = presvote ~ difflog, data = inc.sub)
```

Coefficients:

|             |         |
|-------------|---------|
| (Intercept) | difflog |
| 0.50758     | 0.02384 |

2. Make a scatterplot of the two variables and add the regression line.

```
1 ggplot(inc.sub, aes(difflog, presvote)) +  
2   geom_point() +  
3   labs(title = "Difference in Spending and Voteshare of Presidential  
4     Candidate",  
5         x = "Difference Between Incumbent and Challenger's Spending",  
6         y = "Vote Share of the Incumbent's Party's Candidate") +  
7   geom_smooth(method = lm)
```



3. Save the residuals of the model in a separate object.

```
1 regression2 <- lm(presvote ~ difflog, inc.sub)
2 residuals2 <- regression2$residuals
```

4. Write the prediction equation.

The numbers representing the slope and y-intercept are taken from the linear regression run in R.

$$\hat{Y} = .02384X + 0.50758$$

In the equation, Y-hat represents the predicted value of vote share of the presidential candidate of the incumbent party, and X represents the difference between incumbent's and challenger's spending.

## Question 3

We are interested in knowing how the vote share of the presidential candidate of the incumbent's party is associated with the incumbent's electoral success.

1. Run a regression where the outcome variable is **voteshare** and the explanatory variable is **presvote**.

```
1 lm(voteshare ~ presvote, inc.sub)
```

Call:

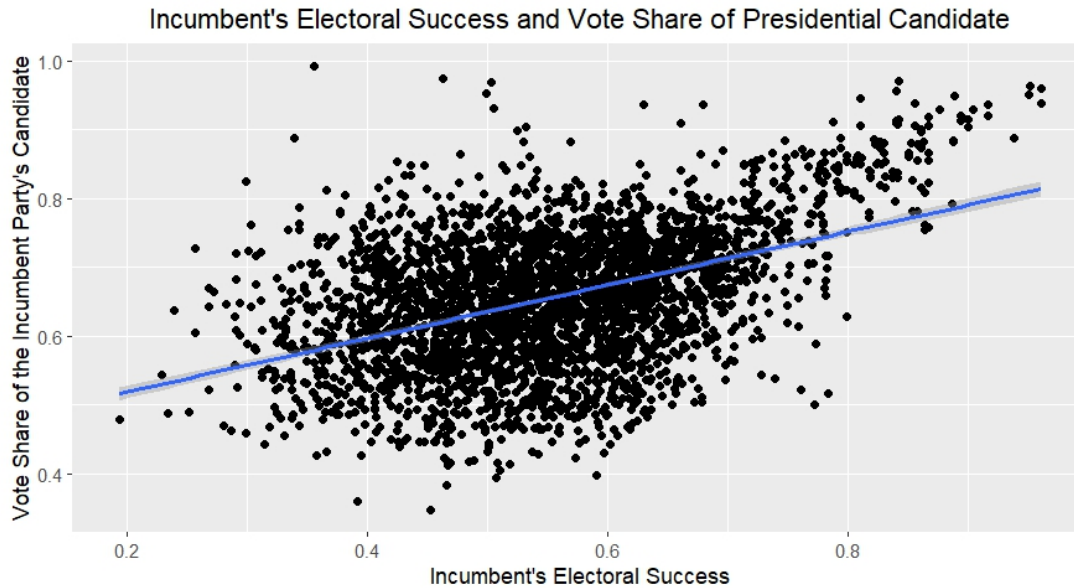
```
lm(formula = voteshare ~ presvote, data = inc.sub)
```

Coefficients:

| (Intercept) | presvote |
|-------------|----------|
| 0.4413      | 0.3880   |

2. Make a scatterplot of the two variables and add the regression line.

```
1 ggplot(inc.sub, aes(presvote, voteshare)) +
2   geom_point() +
3   labs(title = "Incumbent's Electoral Success and Vote Share of
4     Presidential Candidate",
5     x = "Incumbent's Electoral Success",
6     y = "Vote Share of the Incumbent Party's Candidate"
7   ) +
8   geom_smooth(method = lm)
```



3. Write the prediction equation.

The numbers representing the slope and y-intercept are taken from the linear regression run in R.

$$\hat{Y} = 0.3880X + 0.4413$$

In the equation, Y-hat represents the predicted value of the vote share of the president candidate of the incumbent party, and X represents the incumbent's electoral success.

## Question 4

The residuals from part (a) tell us how much of the variation in `voteshare` is *not* explained by the difference in spending between incumbent and challenger. The residuals in part (b) tell us how much of the variation in `presvote` is *not* explained by the difference in spending between incumbent and challenger in the district.

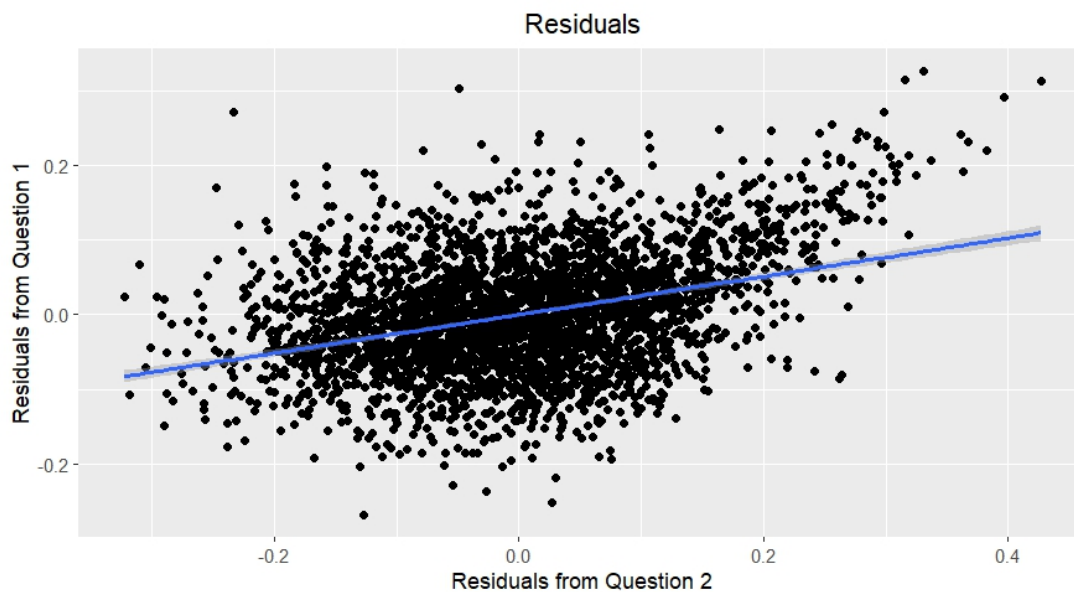
1. Run a regression where the outcome variable is the residuals from Question 1 and the explanatory variable is the residuals from Question 2.

```
1 lm(residuals1 ~ residuals2, inc.sub)
```

```
Call:
lm(formula = residuals1 ~ residuals2, data = inc.sub)
```

```
Coefficients:
(Intercept)          residuals2
-0.00000000000000005934    0.256877012700097440145
```

2. Make a scatterplot of the two residuals and add the regression line.



3. Write the prediction equation.

$$\hat{Y} = 0.256877012700097440145X - 0.00000000000000005934$$

In the equation, Y-hat represents the predicted value of the variation in voteshare that is not explained by the difference in spending between incumbent and challenger and X represents the variation in presvote not explained by the difference in spending between incumbent and challenger in the district.

## Question 5

What if the incumbent's vote share is affected by both the president's popularity and the difference in spending between incumbent and challenger?

1. Run a regression where the outcome variable is the incumbent's `voteshare` and the explanatory variables are `difflog` and `presvote`.

```
1 lm(voteshare ~ difflog + presvote, inc.sub)
```

Call:

```
lm(formula = voteshare ~ difflog + presvote, data = inc.sub)
```

Coefficients:

| (Intercept) | difflog | presvote |
|-------------|---------|----------|
| 0.44864     | 0.03554 | 0.25688  |

2. Write the prediction equation.

The numbers representing the slope and y-intercept are taken from the linear regression run in R.

$$\hat{Y} = 0.44864 + 0.03554X_1 + 0.25688X_2$$

In the prediction equation,  $\hat{Y}$  represents the predicted value of the incumbent's vote share,  $X_1$  represents the difference in spending between incumbent and challenger, and  $X_2$  represents the president's popularity.

3. What is it in this output that is identical to the output in Question 4? Why do you think this is the case?

The residuals from Question 1 represent the part of `voteshare` that cannot be linearly explained by `difflog`. The residuals from Question 2 represent the `presvote` that cannot be linearly explained by `difflog`. In Question 4, we are looking for a linear relationship between the 2 residuals and we got the coefficient 0.25688, which represents the impact of `presvote` on vote share if you take out the affect of `difflog` on `presvote` and take out the impact of `difflog` on vote share. In Question 5, 0.25688 represents the impact of `presvote` on `voteshare`, assuming that `difflog` and `presvote` are uncorrelated. Therefore, since in both cases the variable represents the impact of `presvote` on `voteshare` with no impact of `difflog`, the coefficients are the same. Following this logic, if I were to do a summary of the linear regressions, it would show that the standard errors are also the same.